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Original Article

Correlation between Harris hip score and gait analysis through artificial intelligence pose estimation in patients after total hip arthroplasty

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A R T I C L E I N F O

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A B S T R A C T

Background: Recently, open pose estimation using artificial intelligence (AI) has enabled the analysis of time series of human movements through digital video inputs. Analyzing a person's actual movement as a digitized image would give objectivity in evaluating a person's physical function. In the present study, we investigated the relationship of AI camera-based open pose estimation with Harris Hip Score (HHS) developed for patient-reported outcome (PRO) of hip joint function.
Method: HHS evaluation and pose estimation using AI camera were performed for a total of 56 patients after total hip arthroplasty in Gyeongsang National University Hospital. Joint angles and gait parameters were analyzed by extracting joint points from time-series data of the patient's movements. A total of 65 parameters were from raw data of the lower extremity. Principal component analysis (PCA) was used to find main parameters. K-means cluster, X-squared test, Random forest, and mean decrease Gini (MDG) graph were also applied.
Results: The train model showed 75% prediction accuracy and the test model showed 81.8% reality prediction accuracy in Random forest. “Anklerang_max”, “kneeankle_diff”, and “anklerang_rl” showed the top 3 Gini importance score in the Mean Decrease Gini (MDG) graph.
Conclusion: The present study shows that pose estimation data using AI camera is related to HHS by presenting associated gait parameters. In addition, our results suggest that ankle angle associated parameters could be key factors of gait analysis in patients who undergo total hip arthroplasty.

1. Introduction

Many joint and postoperative functional evaluation tools have been developed and used in orthopedic surgery.[1–3] There are many different types of measurements, including patient-reported, imaging, and laboratory testing.[4] Patient-reported outcome (PRO) is a method of gathering information appraised by patient without intervention of external perspectives.[5] PRO simply means that patients self-record their health status. PRO including well-validated self-reported questionnaires can help diagnosis. It is widely used in orthopedic research and clinical practice.[6] However, PRO is controversial in its objectivity and reliability because it depends on the subjectivity of patients.[7,8] Especially, objective function evaluation and baseline functional status before injury are very important for determining orthopedic research and surgery prognosis.[9] Therefore, an actual gait measurement for physical function evaluation is required in orthopedics.[10–12]

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The Harris hip score (HHS) is a validated tool to measure the hip pathology and functional evaluation of the patient.[13] In particular, HHS is a frequently used and traditional method to evaluate hip joint function before and after hip surgical procedures such as total hip arthroplasty.[14] If a patient's results suggest ongoing hip dysfunction, the Harris Hip Score can be used to determine if they would benefit from additional treatment, such as physical therapy or revision surgery.[15] Therefore, HHS is an important clinical tool because it offers an objective, standardized measurement to assess patients' hip function and monitor their development over time.

Recently, artificial intelligence (AI) has been tried in various medical fields, and many benefits have been confirmed.[16,17] With the development of computers and AI technologies, efforts to automatically analyze gait are also increasing.[18–22] Open pose estimation has enabled the analysis of time series of human movements through digital video inputs.[23–25] Human pose estimation (HPE) offers the location of the human body's representative position from input data.[26] HPE has the advantage of being able to analyze images related to human joints or gait as time-series data.[27] Analyzing actual movement of human by digitized image would give objectivity in evaluating a person's physical function.

In the present study, we investigated the relationship of AI camera-based open pose estimation with the HHS developed for PRO of hip joint function.[28] The gait of patients who underwent total hip arthroplasty was measured with an AI pose estimation. The relationship between gait variables and HHS was then evaluated.

2. Methods

2.1. General information

A total of 65 participants were included in the present study. Participants were hip joint disease patients who were treated in Gyeongsang National University Hospital. This study was approved by the Institutional Review Board (IRB) of Gyeongsang National University Hospital (2022-01-032) and all methods were performed in accordance with the relevant guidelines and regulations. This research was carried out following the ethical protocol. Among those 65 participants, those with missing values (e.g., when frontal or lateral images were not recognized) were excluded. Finally, pose-estimation data of 51 subjects were collected. We obtained informed consent of all participants about follow-up observation and participating in their gait imaging. In addition, informed consent from the subject for publication of identifying images was obtained. Inclusion criteria were as follows (I) patients with a history of diagnosis of hip joint disease; (II) patients with a history of hip replacement arthroplasty; and (III) patients who could walk independently. Exclusion criteria were as follows (I) Patients whose gait film was not properly recorded; (II) Patients who needed support with an instrument or other person while walking. There was no age limitation for participants in this study.

2.2. Rating criteria of HHS

The hip function of patients was evaluated by HHS. The criteria and scoring method followed the modified HHS.[29] Specific questions about pain level, gait symptom of limp, whether assistive devices were required, walking distance, how to go up and down stairs, independent wearing socks or shoes, sitting, public transportation were used as shown in Table 1.

2.3. Gait video and data processing

Gait analysis was conducted using an AI camera (OBSBOT-2007-C, Remo Tech, China). Video data to evaluate patient's movement in time series were extracted. Videos were taken from both front and side views. They were taken according to the following criteria (I) when shooting from the front, the camera was set at a distance of 2.5 m and a height of 0.9 m; (II) when shooting from the side, the camera was set at a distance of 2 m and a height of 130 cm. An overview of video recording for pose estimation is shown in Fig. 1.

All raw data were collected with the application of Dr. Logs (Deevo, South Korea). A data management system (DMS, Deevo, South Korea) was used as a data analysis and collection tool. Video data were processed using Mediapipe Pose, one of the open-source machine learning algorithms for pose tracking. Mediapipe Pose is a high-fidelity body pose tracker that uses RGB video frames to infer 33 3D landmarks and a background segmentation mask for the whole body.[30] It utilizes a two-step detector-tracker pipeline. In the first step, a person/pose region-of-interest (ROI) was placed in a video frame using a detector. In the second step, the tracker predicted pose landmarks and a segmentation mask within the ROI using the ROI-cropped frame as input.[30]

2.4. Pose estimation parameters and HHS grade

Joint angles and gait parameters were analyzed by extracting joint points from time-series data of subject's movements. A total of 65 parameters including gait speed, cadence, knee angle, ankle angle, hip joint angle, and location information at five joint points (hip, knee, ankle, foot, heel) were generated from raw data of the lower extremity. Principal component (PC) analysis was performed to find significant variables of 65 gait parameters. After that, k-means cluster was applied to the time-series data for clustering participants. A chi-square test was performed to determine the difference between clusters and HHS grouped by grade. Grouping of HHS by grade was as follows: very bad (HHS 1–31), bad (HHS 31–55), good (HHS 56–75), and very good (HHS 76–100).

In addition, random forest was performed to determine whether parameters measured in the pose estimation divided participants well by HHS grade. A mean decrease Gini (MDG) graph was also applied to find out which parameters well divided participants by HHS grade.[31]

2.5. Statistics

An unpaired t-test was used for continuous variables to discover significant differences in parameters of groups. The value was expressed as mean ± standard deviation. Significance level was set at $p < 0.05$. For k-means cluster and PCA, ‘kmeans’ and ‘prcomp’ as built-in functions of the R program were used. The chi-square test used R program's built-in function, ‘chisq’. Random forest and MDG analysis were performed using the ‘randomForest’ of R package.

3. Results

3.1. Clustering and identifying relationship between HHS grade and open pose data

PA analysis was performed for parameters obtained through participant's gait imaging. The cumulative proportion was 0.9564 in PC10, which was significant. Thus, subsequent analyses were conducted based on PC10 (Table 2). After that, four clusters (Cluster 1 (n = 16), Cluster 2 (n = 1), Cluster 3 (n = 16), Cluster 4 (n = 18)) were created using k-means clustering. Among them, Cluster 2 had just one participant. Thus, it was excluded because the number of samples was insufficient.

Pearson's Chi-square test was performed to analyze correlation between open pose data and HHS grade. Pearson's chi-square test

Table 1
Harris Hip Score questionnaire.

Question	Score
1. Pain level	
A. Absent or negligible	44
B. Sometimes slight pain, but it does not interfere with daily life	40
C. Mild pain, but it does not interfere with daily activities. May require a mild pain reliever (aspirin)	30
D. Moderate pain, and it does interfere with daily life. Stronger pain reliever may be required.	20
E. Severe pain, and it is difficult to carry out daily life.	10
F. Can only stay in bed due to pain	0
2. Gait – symptom of limp	
A. None at all	11
B. Slightly present	8
C. Moderately present	5
D. Severe	2
E. Unable to walk	0
3. Gait – whether or not assistive devices are required	
A. No	11
B. Use a cane when going far	9
C. Usually use a cane	7
D. Use of crutches on one side	5
E. Use of canes on both sides	3
F. Use of crutches on both sides	1
G. Unable to walk	0
4. Waliking distance	
A. No limit	11
B. 1 km	8
C. 500 m	5
D. Indoor living only	2
E. Unable to walk	0
5. How to go up and down stairs	
A. Available without holding the handrail	4
B. Available by holding the handrail	2
C. Difficult to use stairs	1
D. Not available to use stairs	0
6. Independent wearing socks or shoes	
A. Easy	4
B. Difficult but possible	2
C. Not available	0
7. Sitting	
A. Able to sit on a general chair for about an hour	5
B. Able to sit on a high chair for 30 min	3
C. Unable to sit on a chair	0
8. Public transportation	
A. Available	1
B. Not available	0

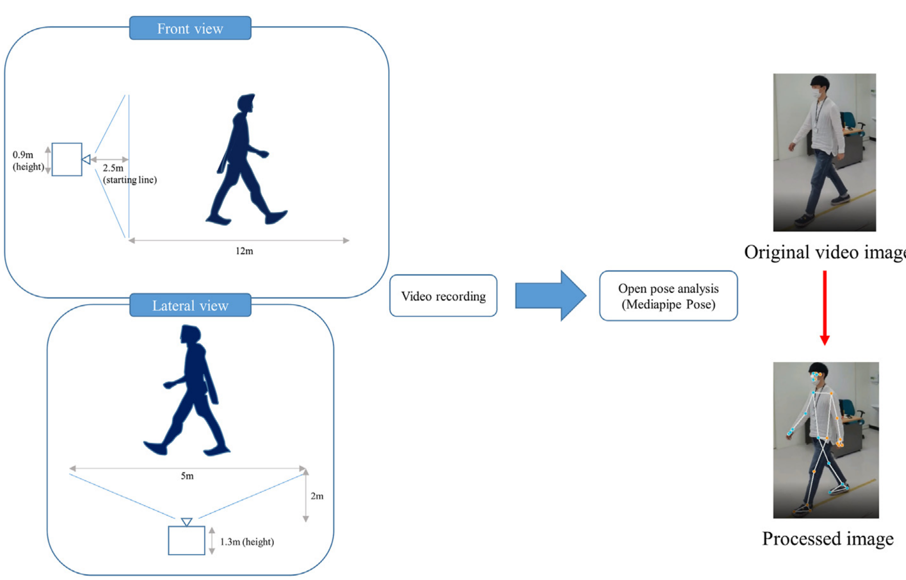

Fig. 1. Overview of video recording for pose estimation.

Table 2
Importance of components in principle component analysis.

PC number	Standard deviation	Proportion of Variance	Cumulative proportion
PC1	23.3532	0.2712	0.2712
PC2	22.2480	0.2462	0.5174
PC3	16.7072	0.1388	0.6562
PC4	14.1335	0.0993	0.7555
PC5	11.1357	0.0617	0.8172
PC6	9.7948	0.0477	0.8649
PC7	8.2541	0.0339	0.8988
PC8	7.0946	0.0250	0.9238
PC9	5.9362	0.0175	0.9413
PC10	5.5091	0.0151	0.9564
PC11	4.7407	0.0112	0.9676
PC12	3.8592	0.0074	0.9750
PC13	3.6575	0.0067	0.9817
PC14	3.1895	0.0051	0.9867

between cluster 1, cluster 3, and cluster 4 showed a significant result ($p = 0.04848$), meaning that clustering through open pose data reflected HHS grade (Table 3).

3.2. HHS grade prediction using pose estimation parameters

To make HHS grade prediction using pose estimation parameters, participants were grouped into HHS grades (very bad, bad, good, very bad). A random forest was then performed for the HHS grade prediction model using pose estimation parameters. These applied pose estimation parameters were extracted from exploratory data analysis of PCA results. A total of 40 samples were used for the training model. Eleven samples were used for the test model for training and testing the random forest machine learning algorithm. As a result, the training model showed a prediction accuracy of 75% (Fig. 2A) and the test model showed a reality prediction accuracy of 81.8% (Fig. 2B). Random forest considers randomly selected variables, rather than considering all variables in the data for each node of every decision tree. Due to the nature of this algorithm, it was not possible to know which variable was selected for each node, making it difficult to explain which variable used affected characteristics of the result presented as a random forest. Therefore, specific open pose parameters that could classify HHS grade were analyzed through MDG graph. As a result, “anklerang_max” meaning the maximum value of the ankle angle, “kneeankle_diff” meaning the value obtained by subtracting the sum of the maximum values of the knee and ankle angles by the sum of the knee and ankle angles when standing, and “anklerang_rl” meaning the difference between the left and right ankle angles showed the top 3 Gini importance score (Fig. 3).

4. Discussion

The Harris Hip Score (HHS) is a validated tool to assess a person's functional capacity. It has been the most common scoring method used to evaluate the condition of a patient's hip pathology.[15] It has also been used to evaluate the outcome of total hip replacements (THRs) in many studies. HHS has been used to evaluate results of THRs. A modified version of HHS that removes the highly variable physical examination component has been developed. It is used as a more reliable indicator.[13,32–34] As a result, HHS is evolving into a less medically specialized and simpler version. It can be measured without having direct contact with patient.[35]

Parameters measured in gait analysis data in the present study classified people and groups reflecting HHS grade. In addition, we verified the HHS grade prediction model using parameters extracted from gait analysis of open pose using random forest, one of the machine learning techniques. As a result, the train model showed a prediction accuracy of 75% and the test model showed a reality prediction accuracy of 81.8%. According to the study of Shahadat Uddin comparing different machine learning algorithms for disease prediction, Random Forest algorithm showed superior accuracy comparatively.[36] Specific open pose parameters that could classify HHS grade were also analyzed through MDG graph. It was found that anklerang_max, kneeankle_dif, and anklerang_rl showed the top 3 Gini importance score. These are variables commonly associated with ankle angle. The ankle angle is a key factor that affects the gait cycle according to the maximum value or symmetry of the angle. It is important to evaluate variables of the ankle joint when studying gait kinematics.[37,38] Therefore, this study suggests that the ankle angle can be a potential factor for gait evaluation through open pose by showing that classification of patients who have undergone hip arthroplasty through the ankle angle reflects the HHS grade.

The present study showed that pose estimation data using AI camera were related to HHS by presenting associated gait parameters. In addition, our results suggested that ankle angle associated parameters could be key factors of gait analysis in patients who underwent total hip arthroplasty. Therefore, pose estimation data using AI camera could be a potential indicator for functional evaluation after total hip arthroplasty. Further study is needed to compare evaluation methods for various gait-related diseases and open pose data.

Table 3
Chi-square test for determining differences between clusters and Harris Hip Score grouped by grade.

Cluster	HHS grade			
	Very bad	Bad	Good	Very good
Cluster 1 (n = 16)	0	6	6	4
Cluster 3 (n = 16)	0	3	7	6
Cluster 4 (n = 18)	4	7	6	1

X-squared = 12.676, df = 6, p-value = 0.04848.

5. Limitation

First, our open pose data did not consider all variables. Therefore, further study is needed to determine which combination of parameters could best reflect HHS. Second, the number of participants in each group was small. Comparing more gait images will be able to present more reliable data. Third, the group of patients with open pose data was not uniform. HHS is usually performed to assess hip function after hip or extremity-related surgery. However,

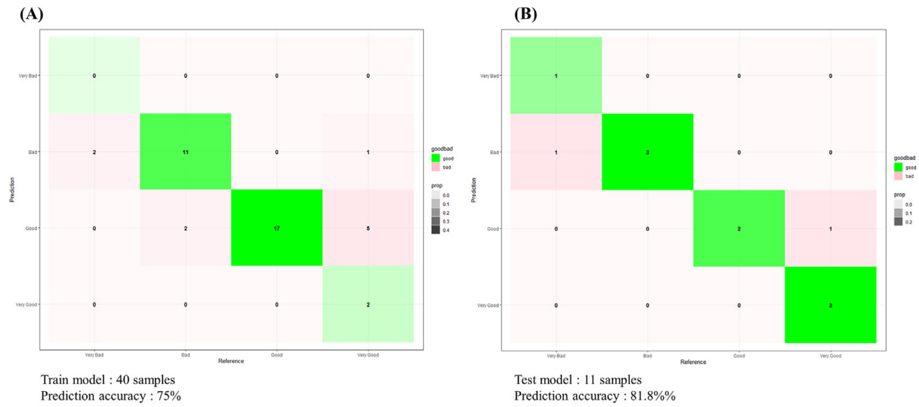

Fig. 2. Confusion matrix of random forest for Harris Hip Score grade prediction model using pose estimation parameters. (A) Train model, (B) Test model.

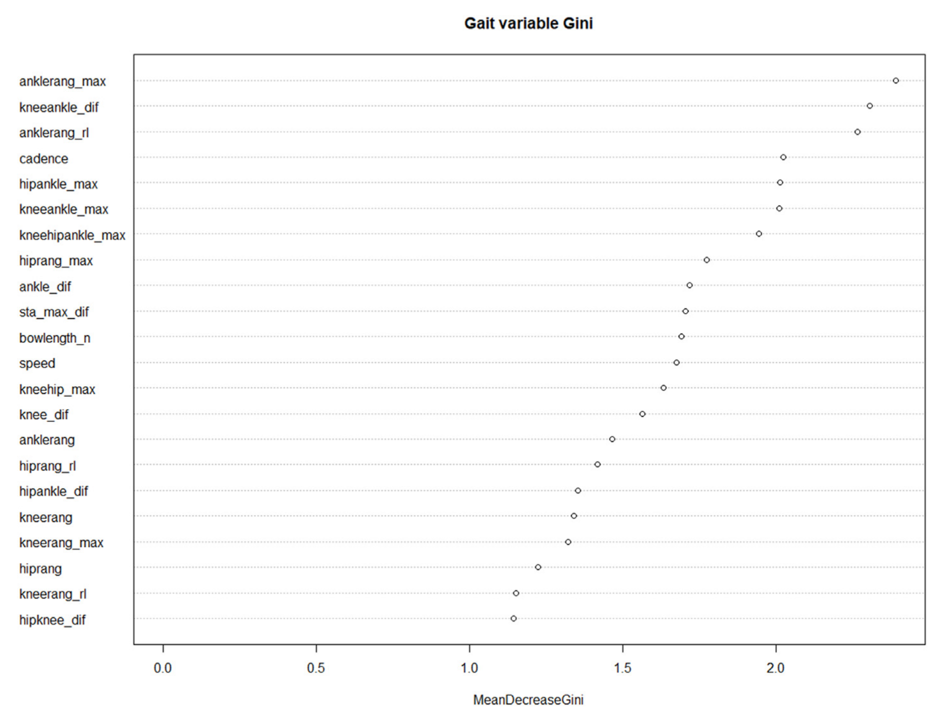

Fig. 3. Mean decrease Gini graph of gait parameters in R.

participants' open pose evaluation time were inconsistent. Therefore, open pose data analyses for large and well-refined populations are needed in the future.

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Nothing to declare.

Ethical approval

This study was approved by the Institutional Review Board (IRB) of Gyeongsang National University Hospital (2022-01-032) and all methods were performed in accordance with the relevant guidelines and regulations.

Informed consent

We obtained informed consent of all participants about follow-up observation and participating in their gait imaging. In addition, informed consent from the subject for publication of identifying images was obtained.

Consent for publication

All authors agree for publication.

Availability of data and material

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declaration of competing interest

None declared by all authors.

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